

IS-LM Equilibrium

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Introduction

In this section, we build the short-run IS-LM model

You will learn:

1. how to set up and interpret the IS-LM model
2. what its limitations are
3. how to solve for the equilibrium
4. how to analyze the effects of shocks and policies

The IS/LM model is mostly used as a building block for the AS/AD model.

All of this is review of material you previously learned in Intermediate Macro.

IS-LM Model

Key assumptions:

- Output is determined by aggregate demand
- There is no supply side
- Prices are fixed
- Closed economy

Think: economy in recession, with lots of unemployed resources.

We relax all of these assumptions later.

I IS-LM Model

I.I Model Setup

Two markets

- Goods (IS). Money (LM)
- In the background there is also a bond market

Two endogenous variables

- Output (Y). Interest rate (i)

Two policy variables

- Government spending (G). Money supply (M)

I.2 Modeling Choices

Why does the model have these ingredients?

We want to talk about monetary and fiscal policies

- main instruments for business cycle stabilization

Main outcome of interest: output Y

- so we need aggregate demand for goods

Monetary policy mainly works by changing interest rates i

- so we need the money / bond market

Fixed prices are for simplicity only.

I.3 Modeling Choices

The IS/LM model is the **simplest** model in which we can talk about

- monetary policy
- fiscal policy

A natural reaction is: “The model is too simple. It lacks feature X.”

What makes a good model?

- What are models for?

I.4 What do models do?

Models are stories told in math.

It's often a good idea to start really simple.

Tell the story as clearly as possible.

(You can always add bells and whistles later.)

I.5 The IS/LM story

The IS/LM is about **getting economies out of recessions**.

The government has three ways of affecting AD

1. Buy goods (G)
2. Cut taxes (T) because AD depends on after tax income.
3. Change interest rates (i) because the interest rate affects saving (S) and borrowing for investment (I)

There are complicated interactions because

- i depends on AD (money market clearing)
- AD depends on income (a multiplier effect)

2 The Goods Market: IS Curve

2.1 Aggregate demand

AD keeps track of how demand depends on government spending, taxes, and interest rates.

Start from an accounting identity

$$Z = C + I + G + \underbrace{X - IM}_{\text{Net exports}}$$

Z is aggregate demand / expenditure.

For now: closed economy with $X - IM = 0$.

Add behavioral assumptions to give it content.

2.2 Consumption function

What does consumption depend on?

- current income (not as obvious as it looks...)
- interest rates
- expected future incomes
- wealth, ...

That's complicated ...

But we are mainly interested in

- current income (multipliers)
- interest rates (monetary policy)

2.2 Consumption function

So let's assume

$$C = C(Y_D) = c_0 + c_1 Y_D$$

$Y_D = Y - T$: disposable income (after taxes and transfers)

c_0 : “autonomous consumption” (intercept)

c_1 : marginal propensity to consume (slope)

$s = 1 - c_1$: marginal propensity to save

All other determinants of C are stuffed into c_0 (not our main focus).

2.3 Investment function

What does investment depend on?

- interest rate i : cost of capital
- output Y : aggregate demand
- expectations of future business conditions, current capital stock, ...

Also complicated ...

But we focus on Y and i and assume

$$I = I(Y, i) = \bar{I} + b_1 Y - b_2 i$$

Everything else is stuffed into the intercept $\bar{I}$

2.4 Government

We treat G and T as **exogenous**.

- G is government consumption
(not government spending on transfers, salaries, etc)
- T is tax revenue net of transfer payments

Exogenous vs endogenous variables ...

2.5 Aggregate Demand

$$\begin{aligned} Z &= C + I + G \\ &= \underbrace{c_0 + c_1(Y - T)}_C + \underbrace{\bar{I} + b_1Y - b_2i}_I + G \\ &= \underbrace{c_0 - c_1T + \bar{I} + G}_{\bar{Z}} + (c_1 + b_1)Y - b_2i \end{aligned}$$

$\bar{Z}$: autonomous spending / demand

- collects all the intercept terms

In words / intuition ...

2.6 Isn't this completely arbitrary?

Yes, it is arbitrary.

- We just made up a consumption function from introspection.
- When the model was developed, the choices were based on data.

Is this “bad”?

- It depends on the purpose of the model.

A model is a story told through math.

- This model tells the IS/LM story (see above).

2.7 Where are the prices?

$$Z = \bar{Z} + (c_1 + b_1)Y - b_2i$$

Shouldn't a demand curve be downward-sloping in prices?

IS/LM assumes that **prices are fixed**.

- A major limitation.
- Inspired by the Great Depression.

3 Market clearing

3.1 Goods Market Clearing

Output equals demand:

$$Y = Z = \bar{Z} + (c_1 + b_1)Y - b_2i$$

Solve for the level of output that clears the goods market:

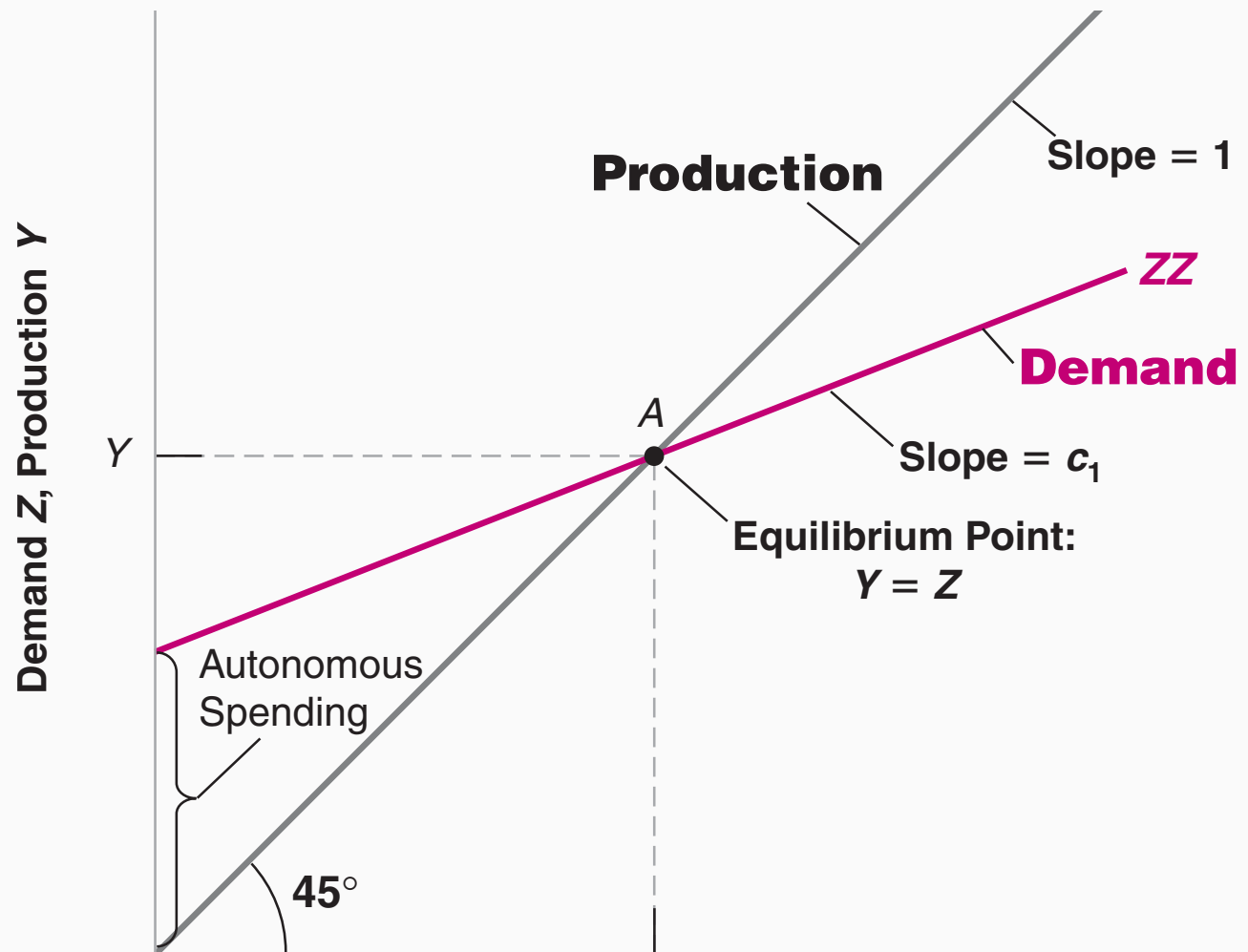
$$Y - (c_1 + b_1)Y = \bar{Z} - b_2i$$

$$Y = \frac{\bar{Z} - b_2i}{1 - c_1 - b_1}$$

Key assumption: marginal propensity to spend $c_1 + b_1 < 1$.

- otherwise we have serious problems...

3.2 Goods Market Clearing



Supply: $Y = Z$

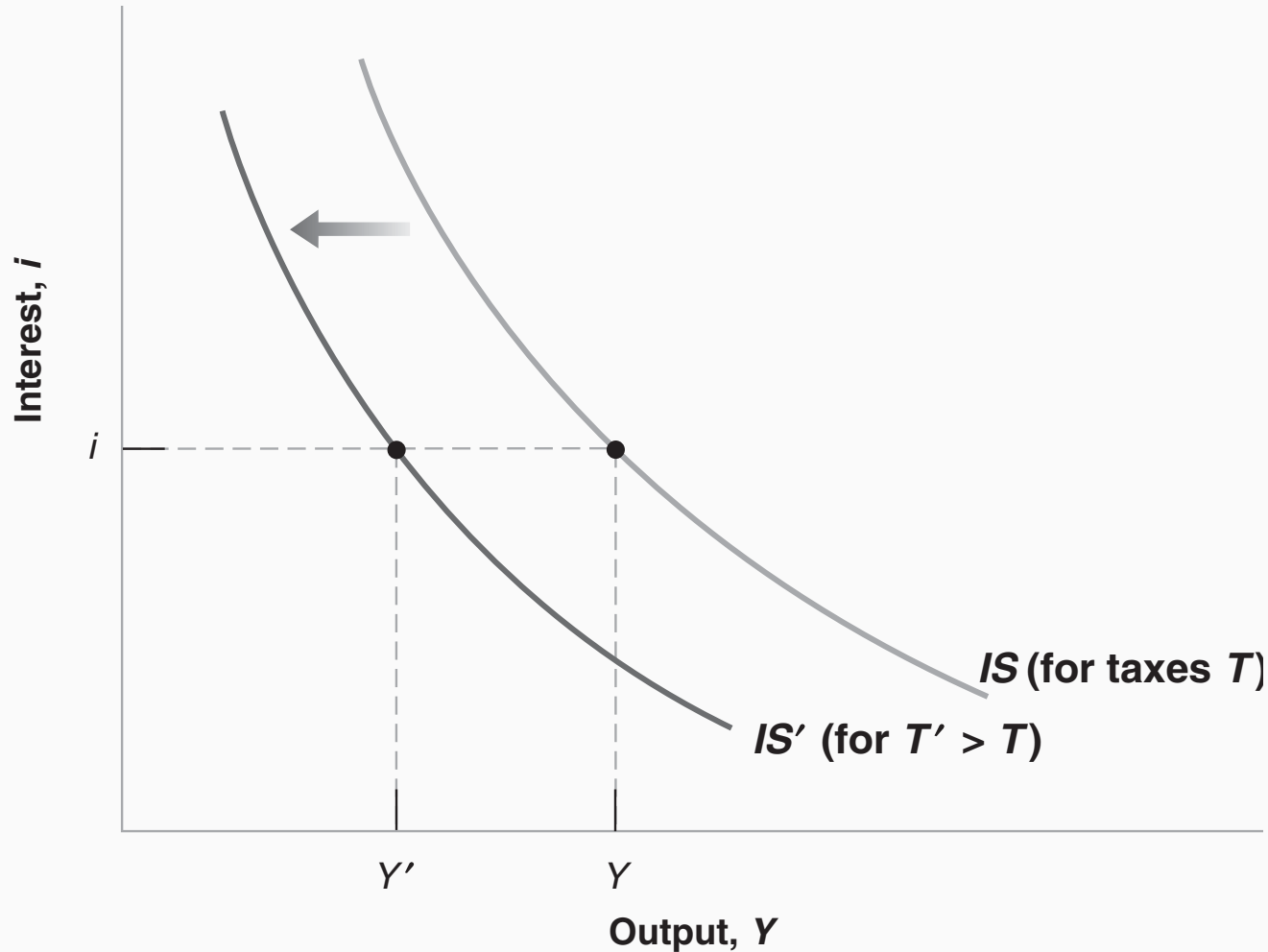
- 45 degree line

Demand:

$$Z = \bar{Z} + (c_1 + b_1)Y - b_2i.$$

What happens when the interest rate i rises?

3.3 IS Curve



IS collects all (Y, i) for which the goods market clears.

3.4 IS equation

We derived

$$Y - (c_1 + b_1)Y = \bar{Z} - b_2i$$

IS plots i against Y .

So we have to solve for i :

$$Y(1 - c_1 - b_1) - \bar{Z} = -b_2i$$

IS equation:

$$i = \frac{\bar{Z} - (1 - b_1 - c_1)Y}{b_2}$$

3.5 Intuition: IS Curve

IS downward sloping:

$$i = \frac{\bar{Z} - (1 - b_1 - c_1)Y}{b_2}$$

Intuition ...

3.6 Shifting the IS Curve

$$i = \frac{\bar{Z} - (1 - b_1 - c_1)Y}{b_2}$$

Y and i are on the axes – they don't shift the curve.

Everything else that appears in the IS equation does.

- Higher $\bar{Z}$ shifts IS up / right. (Intuition?)
- Changing b_1, b_2, c_1 alters the slope (what is the slope?).

Clearly distinguish moving along the curve vs. shifting the curve!

3.7 The Fiscal Multiplier

$$Y = \frac{\bar{Z} - b_2 i}{1 - c_1 - b_1}$$

\$1 of government spending increases Y by

$$\frac{dY}{dG} = \frac{1}{1 - c_1 - b_1} > 1$$

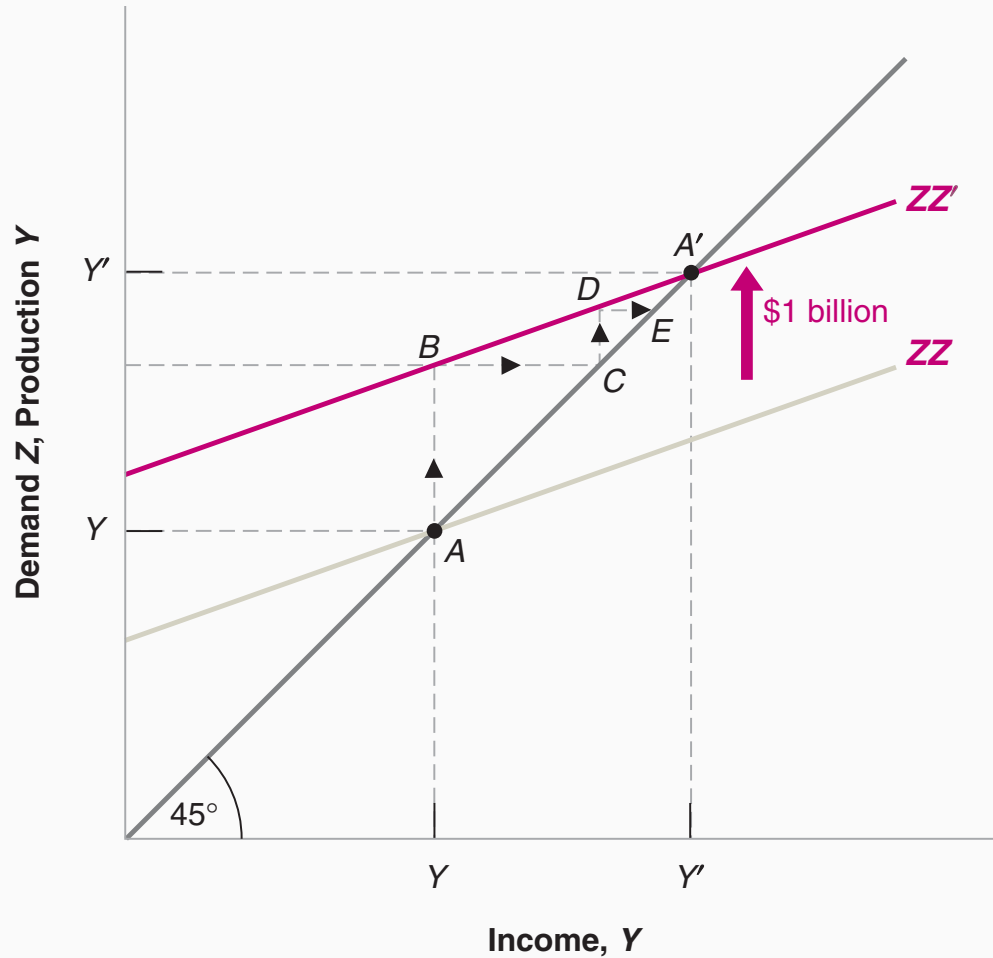
This holds the interest rate constant

- which will not be true in equilibrium

Intuition

- How does a higher MPC (c_1) affect the multiplier?

3.8 The Fiscal Multiplier



3.9 Saving Equals Investment

We can also think about goods market clearing as equating saving with investment.

- Private saving: $S = Y_D - C = Y - T - C$
- Public saving: $S^P = T - G$

Total saving equals investment:

$$I = \underbrace{Y - T - C}_S + \underbrace{T - G}_{S^P}$$

This yields goods market clearing

$$Y = C + I + G$$

3.10 Summary

The IS Curve clears the goods market.

It is downward sloping:

$$i = \frac{\bar{Z} - (1 - b_1 - c_1)Y}{b_2}$$

Government spending $G \uparrow$ and other increases in autonomous demand shift IS right.

3.11 Questions

$$i = \frac{\bar{Z} - (1 - b_1 - c_1)Y}{b_2}$$

1. If taxes T rise, what happens to IS?
2. If investment is highly interest sensitive (high b_2), what happens to IS?
3. If the MPC (c_1) is high, what happens to IS?

4 The Bond Market: LM Curve

4.1 LM Curve

The IS/LM model has two markets:

1. Goods $\rightarrow$ IS curve
2. “Money” $\rightarrow$ LM curve

Two market clearing conditions.

What is “money”?

4.2 Key point

Any definition of money is arbitrary.

In reality, there are assets with higher or lower liquidity.

There is no sharp distinction between money (liquid) and “bonds” (illiquid) assets.

4.3 Money demand Logic

Consider a household with a **fixed amount of wealth**.

The household can hold “money” or bonds.

- Money: liquidity benefit
- Bonds: interest benefit

Division depends on

- transactions volume (nominal income)
 - need for liquidity
- interest rate

4.4 Money Demand

Money demand can then be written as

$$M^d = \$Y \times L(i)$$

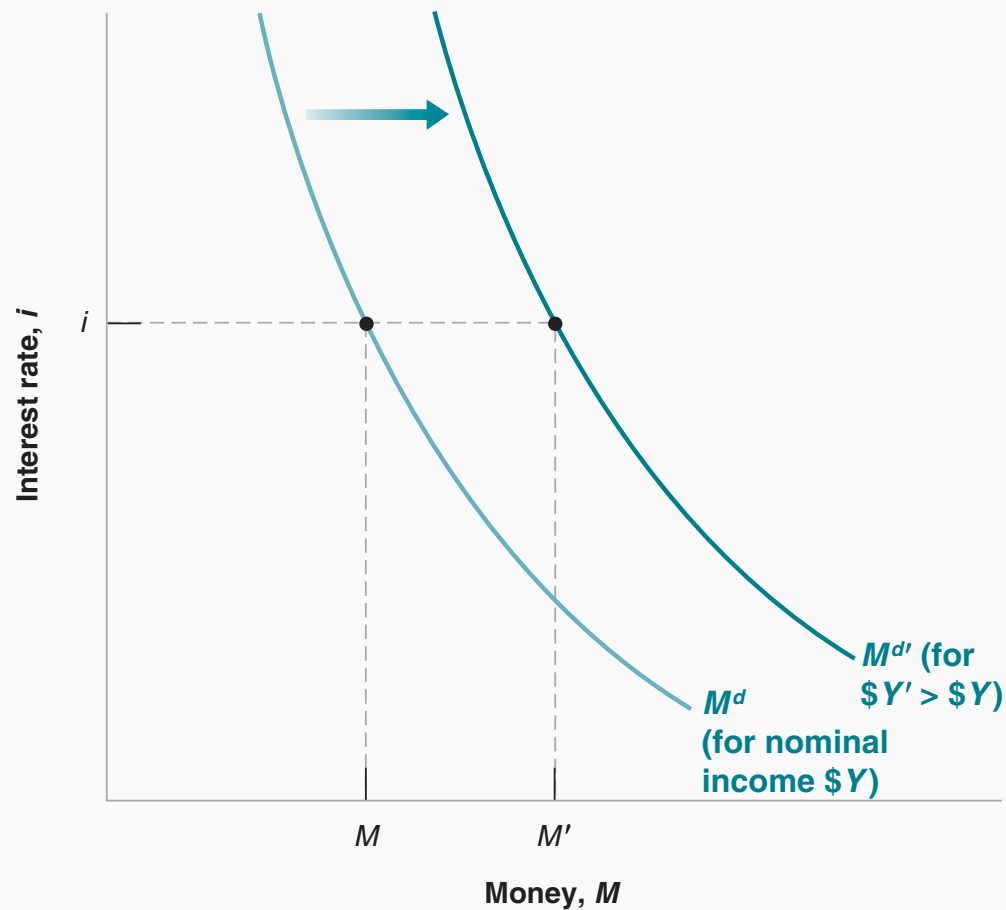
$\$Y$ is nominal income (in dollars)

There are other determinants of money demand

(wealth, inflation expectations, ...)

The IS/LM model focuses on Y and i to tell its story.

4.5 Money Demand



4.6 Money Supply

Real world: money = [currency] + [checkable deposits] + [more?]

Currency: controlled by CB

Checkable deposits: created by banks (**not** controlled by CB)

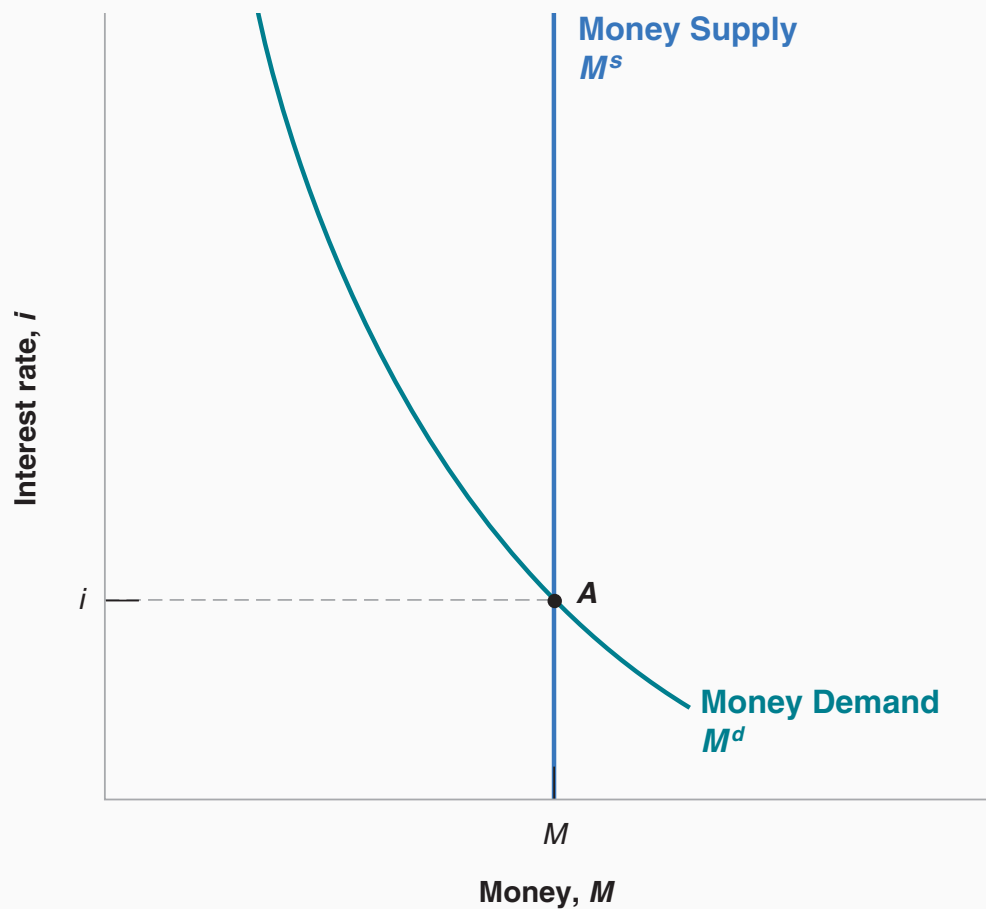
For now: assume that CB controls money supply

$$M = M^s$$

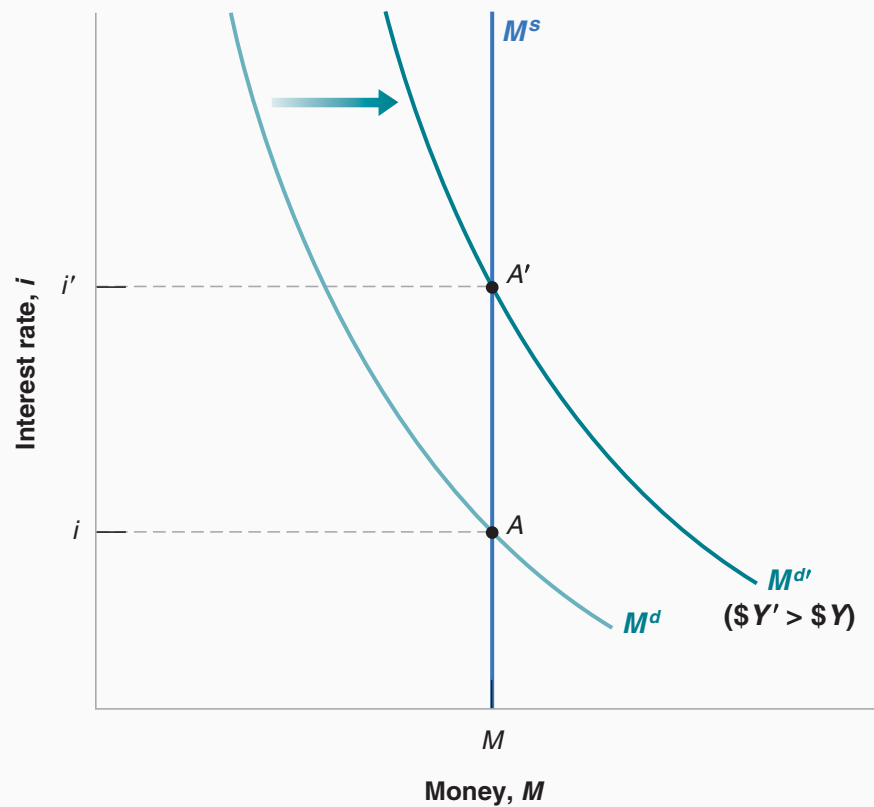
Money market clearing:

$$M^s = \$YL(i)$$

4.7 Money Market Clearing



4.8 Money Demand Increases

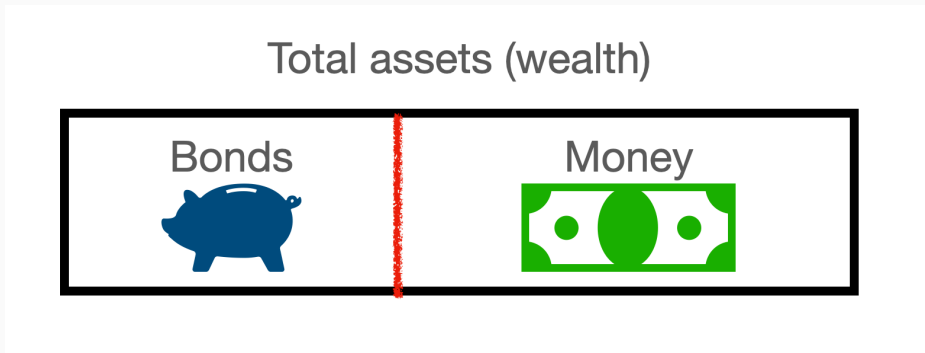


The process...

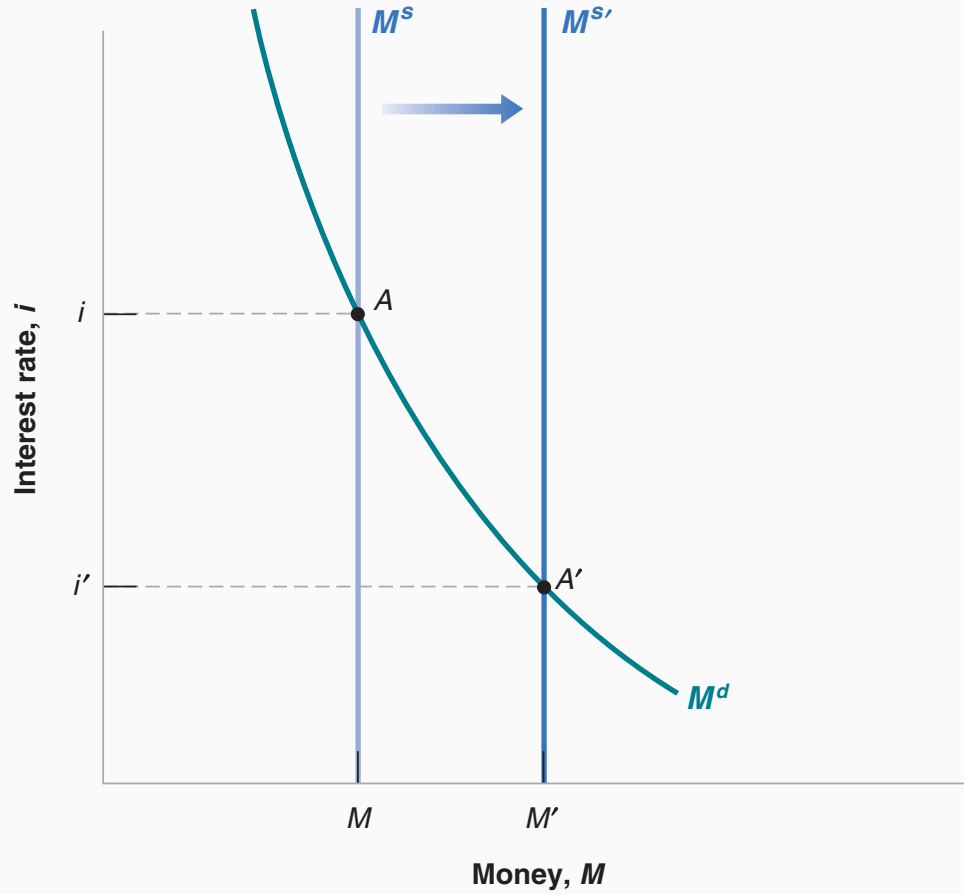
4.9 Money Demand Increases

Key assumption of the model:

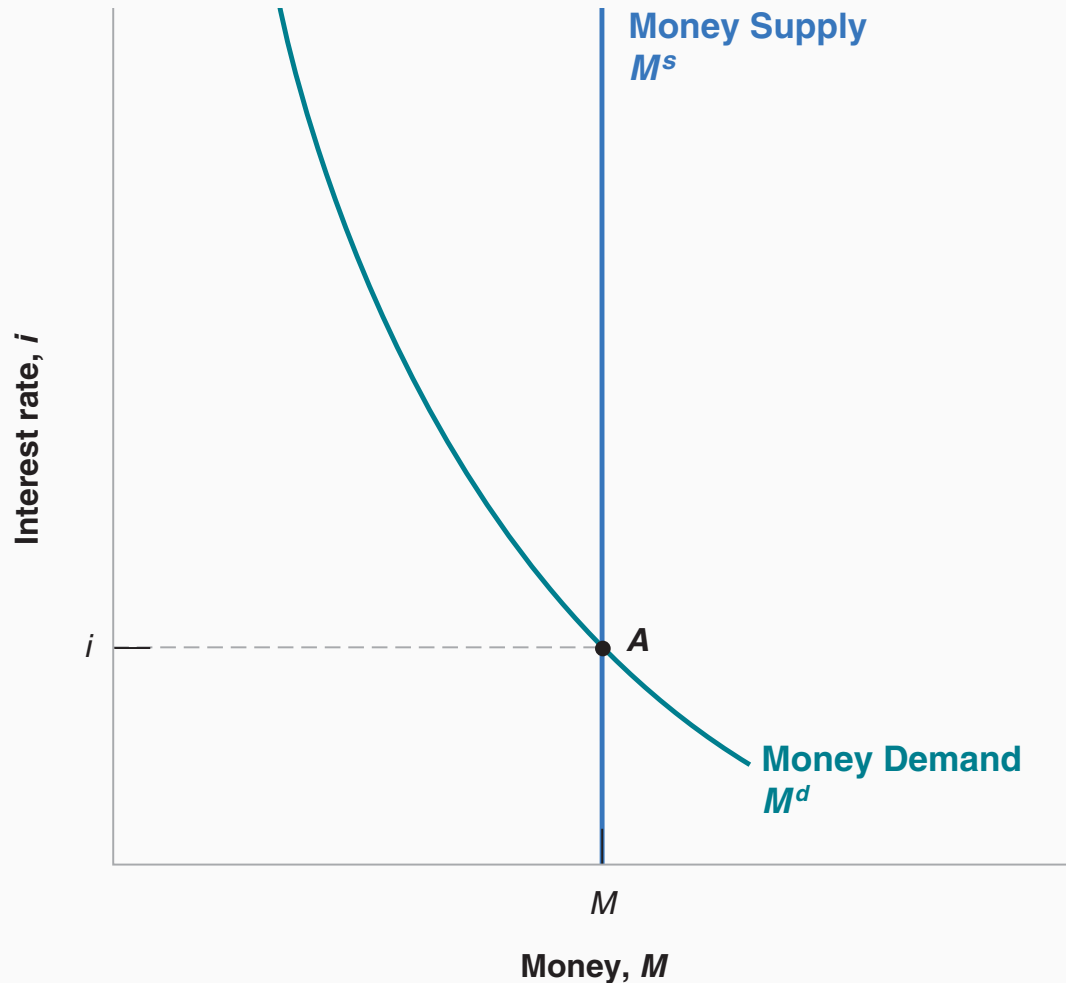
- When i changes, the household can only change the composition of wealth
- Money vs bonds
- Total wealth is fixed.



4.10 Money Supply Increases



4.11 Deriving the LM Curve



Vary Y

(on the x axis in the IS/LM graph).

Trace out i that clears money market.

4.12 Open Market Operations

The markets for money and bonds are linked.

To increase the money supply, the CB buy bonds and pays with currency.

- “open market operation”

The price of bonds rises $\implies$ the bond yield i falls.

- the lower i is what the Fed aims for
- lower i stimulates aggregate demand

A complication:

- the Fed has no direct control over the supply of bonds / the bond interest rate.
- open market operations do not always work

4.13 Summary

We have the ingredients of the IS/LM model:

1. IS curve

$$i = \frac{\bar{Z} - (1 - b_1 - c_1)Y}{b_2}$$

2. LM curve

$$M = \$Y \times L(i)$$

The pair (Y, i) that clears both markets is the equilibrium of the model.

4.14 Reading

- Blanchard / Johnson, Macroeconomics, ch. 3-4